UHMEP API

Referral prescription (RIZIV-INAMI)

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SMALS

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1 Document management

1.1 Document history

Version	Status	Date	Author(s)	Modifications
0.1	Final	25/05/2023	Gaël Mavangui, Julien Beard, Cyprien Janssens	First version of the cookbook for the release 1.0 scope « MVP » of UHMEP
0.2	/		Julien Beard, Cyprien Janssens	Second version, changes are listed in the release note
0.3	Final	26/01/2024	Cyprien Janssens, Julien Beard, Gérard Bigaj	Alignment with the new version in acceptance
0.4	Final	03/07/2024	Cyprien Janssens, Julien Beard, Gérard Bigaj	Acceptance release of June 2024. API changes are listed in the release note.
0.5	Final	04/02/2025	Cyprien Janssens, Julien Beard, Lionel Cremer, Gérard Bigaj	Acceptance release of December 2024.
0.8	Final	25/06/2025	Cyprien Janssens, Julien Beard, Gérard Bigaj	Acceptance release of June 2025
1.0	Final	17/02/2026	Lionel Cremer, Cyprien Janssens, Gérard Bigaj	Production Pilot of February 2026
1.1	Release Note	12/03/2026	Cyprien Janssens	Added improved release note, section 1.2
1.2	Final	07/04/2026	Cyprien Janssens	Release 1.1.1

1.2 Release note

The release note can be found on GitHub.

2 Reference

2.1 eHealth reference

All referenced documents are available on the portal of the eHealth platform. These versions, or any following ones, can be used for the eHealth platform service.

ID	Title	Version	Date	Author
1	eHealth Services – Web Access	2.0	12/07/2018	eHealth platform
2	Identity & Authorization Management (I.AM) Mobile integration	1.7	31/03/2023	eHealth platform
3	Pseudonymization REST	1.0	04/04/2023	eHealth platform
4	Identity & Authorization Management (IAM) eXchange	1.1	28/06/2024	eHealth platform

2.2 FHIR references

FHIR documentation type	Link
Implementation guide	https://build.fhir.org/ig/hl7-be/referral/branches/nursing-pilot/guidance.html
BeReferralServiceRequest	https://build.fhir.org/ig/hl7-be/referral/branches/nursing-pilot/StructureDefinition-be-referral-servicerequest-nursing.html
BeReferralTask	https://build.fhir.org/ig/hl7-be/referral/branches/nursing-pilot/StructureDefinition-be-referral-task.html
BePerformerTask	https://build.fhir.org/ig/hl7-be/referral/branches/nursing-pilot/StructureDefinition-be-performer-task.html
BeOrganizationTask	https://build.fhir.org/ig/hl7-be/referral/branches/nursing-pilot/StructureDefinition-be-organization-task.html
BePractitionerRole	https://build.fhir.org/ig/hl7-be/core//StructureDefinition-be-practitionerrole.html
BePractitioner	https://build.fhir.org/ig/hl7-be/core//StructureDefinition-be-practitioner.html
Examples	https://build.fhir.org/ig/hl7-be/referral/branches/nursing-pilot/StructureDefinition-be-organization-task.html

3 Document information

3.1 Glossary

Term	Definition
Reference ID	This is a unique number assigned by UHMEP to each referral prescription issued.
UHMEP	This is the name given to the API which allows to create and manage referral prescriptions and propositions. UHMEP stands for “Unaddressed Health Message Exchange Platform”.
Therapeutic link	It is a relation that a healthcare professional must establish with the patient to have access to his medical data.
Therapeutic exclusion	This relation can be created by the patient and allows him to prevent a healthcare professional to access to medical data coming from other professionals.
Prescriber	The person who can prescribe a referral prescription for a patient.
Caregiver	The caregiver is a general term for the person who provides care based on the information written on the referral prescription. Also known as "treatment provider" (prescriber as well as treatment provider are considered to be caregivers).
Patient	The individual which is the subject of the referral prescription.
Assignation	The assignation is a recommendation done by the prescriber or is done by the patient to help the caregiver to find the prescription easily.
Execution	The execution represents the status of the care that must be provided by a caregiver.

3.2 Formatting explanation

- When a word is in bold and italic: that is a field in the FHIR specification. Ex: ***status***.
- When a word is between quotations mark “”: This is the value that the specific field can have. Ex: “ready”.
- All FHIR resources mentioned in this document are described in the official eHealth FHIR implementation guide pages. Specific links are available in section 2.2.
- When a role is between parentheses in the begin of sentences, it means that the business rules explained next to it is applicable to this role. Ex: (Prescriber).

4 Goal of the document

This cookbook provides external integrators with the technical specifications required to connect to the UHMEP API. UHMEP uses the FHIR (Fast Healthcare Interoperability Resources) standard to exchange medical data.

The goal of this guide is to take a developer from initial authentication to successfully executing complex operations on referral prescriptions and care proposals.

To help you navigate the integration, the API operations in Section 8 are structured as follows:

- **Description:** The business purpose of the operation.
- **Allowed Roles:** Which user types (e.g., Prescriber, Patient) are authorized to call the endpoint.
- **Endpoint:** The HTTP method and URL path.
- **Implemented Rules:** The specific validation, state, and security rules the API enforces (which may result in 4xx errors if violated).
- **Request & Response:** Expected payload structures, specific FHIR constraints, and expected HTTP status codes.

5 Support

5.1 Report Issues

For issues related to UHMEP:

- Acceptance: integration-support@ehealth.fgov.be.
- Production: support@ehealth.fgov.be

5.2 Certificates

To access the secured eHealth platform environment each integrator must obtain an eHealth platform certificate, used to identify the initiator of the request. Please consult the chapter about the eHealth Certificates on the portal of the eHealth platform:

- https://www.ehealth.fgov.be/ehealthplatform/nl/ehealth-certificaten
- https://www.ehealth.fgov.be/ehealthplatform/fr/certificats-ehealth

For technical issues regarding eHealth platform certificates:

- Acceptance: acceptance-certificates@ehealth.fgov.be
- Production: support@ehealth.fgov.be

6 Global overview

6.1 Context (Goal of the project)

UHMEP ("Unaddressed Health Message Exchange Platform") is a centralized exchange platform that stores non-drug referral prescriptions (e.g., wound care, annex81) and makes them available to healthcare professionals and patients via a RESTful API.

By digitizing this workflow, UHMEP eliminates paper prescriptions, centralizes care data, and standardizes data formats using national FHIR templates defined by the eHealth standardization team.

Referral prescriptions created strictly for internal hospital use fall outside the scope of UHMEP.

6.2 Access management:

6.2.1 eHealth token

All API requests to UHMEP must be authenticated. UHMEP identifies the connected user via an eHealth access token.

To access UHMEP, integrators must use an eHealth access token. Because standard user tokens contain pseudonymization rights (which UHMEP is not authorized to receive), the integrator must perform a **Token Exchange**. The original token is exchanged for a restricted token that grants access to the UHMEP backend. This exchanged token must be passed in the Authorization header of all API requests.

Technical information about token generation and token exchange is available in eHealth pages, see section 2.1.

The exchanged token provides UHMEP with the user's SSIN, discipline (profession type), and role. *(Note: The patient's SSIN will be pseudonymized and present only in the user info token).*

For organizations, the token provides UHMEP with an organization's relevant information such as the presence of a CoT (Circle of Trust) and the organization's NIHDI number.

6.2.2 Roles

A connected user is assigned one role, which act as the first layer of API access control.

- **Patient**: Can consult their own prescriptions, tasks, and lists. Can assign/remove caregivers or organizations, and cancel their own prescriptions.
- **Prescriber**: Can create, consult, and cancel prescriptions. Can consult lists and caregiver execution tasks.
- **Caregiver**: Can consult prescriptions/lists/tasks, execute treatments (start, finish, interrupt, cancel), assign/remove caregivers or organizations, and transfer assignations.
- **Organization**: Can create prescriptions and proposals, consult list/details/execution tasks of prescriptions and proposals, as well as assign them to itself or healthcare professionals part of the organization. Organizations are also able to remove an assignation on a prescription/proposal
  The "Restricted" Role: If a healthcare professional's token only contains the "Restricted" role, it indicates their license is suspended. They will have strictly limited access to UHMEP (see Section 6.2.4).

6.2.3 Access matrix

Beyond the base role, UHMEP enforces an Access Matrix defined by INAMI. Before allowing an action, the API evaluates:

1. The operation requested.
2. The prescription template type (defined by INAMI).
3. The discipline of the professional (retrieved from the token) or of the organization.

If the user's or organization's discipline is not authorized for the requested template/action, the API will return a 403 Forbidden error.

6.2.4 Suspension

If a healthcare professional is suspended, they lose active working privileges. However, they are permitted to retroactively log work in UHMEP that was completed prior to their suspension date. When suspended, certain API operations are completely blocked, while others have modified behaviors. This will be described in the section 8.

Note: In the scope of Machine-to-Machine (M2M), UHMEP API won't be enforcing those verifications since they are each organization's responsibility.

6.2.5 Informed Consent, Therapeutic Relationship & Exclusion

UHMEP strictly enforces patient privacy preferences via Attribute Authority checks. Depending on the endpoint called, the API verifies:

- **Informed Consent:** Whether the patient has given their global informed consent.
- **Therapeutic Relationship:** Whether an active relationship exists between the professional and the patient.
- **Therapeutic Exclusion:** Whether the patient has explicitly blocked this professional from accessing their data.

Note: In the scope of Machine-to-Machine (M2M), UHMEP API won't be enforcing those verifications since they are each organization's responsibility.

Please refer to Annex A for further information.

6.3 FHIR

FHIR (Fast Healthcare Interoperability Resource) is used by UHMEP as the transport layer. All communications with UHMEP should follow guidelines defined by HL7. The Belgian version of FHIR is managed by eHealth. All links in this document should refer to the Belgian FHIR resources when provided, or to international ones if no changes have been enforced in Belgium.

All resources used in UHMEP are defined and described in the implementation guide artifacts from HL7 Belgium. This document will not explain different fields of these resources but will refer to the FHIR profile definition used by UHMEP. In general, in each resource, UHMEP supports all fields marked as "Must Support" and all mandatory fields in the FHIR documentation.

For instance, Figure 1 shows a portion of the implementation guide of the BeReferralTask and it states that:

- ***status*** has a cardinality of "1..1" so it is mandatory and used by UHMEP
- ***statusReason*** and ***intent*** are marked as "Must Support" (the red "S"), so they are used by UHMEP
- ***businessStatus*** is optional and not marked as "Must Support", so it may be discarded by UHMEP ***(This will be confirmed in a future release)***.

status	?! Σ	1..1	code	draft | requested | received | accepted | + **Binding:** TaskStatus (required): The current status of the task.
statusReason	S Σ	0..1	CodeableConcept	Reason for current status **Binding:** BeVSTreatmentStatusReason (required)
businessStatus	Σ	0..1	CodeableConcept	E.g. "Specimen collected", "IV prepped" **Binding:** (unbound) (example): The domain-specific business-contextual sub-state of the task. For example: "Blood drawn", "IV inserted", "Awaiting physician signature", etc.
intent	S Σ	1..1	code	unknown | proposal | plan | order | original-order | reflex-order | filler-order | instance-order | option **Binding:** BeVsRequestIntent (required)
priority		0..1	code	routine | urgent | asap | stat **Binding:** RequestPriority (required): The task's priority.

Figure 1: Part of the profile definition of the BeReferralTask

The implementation guide is a guide that should be used by both integrators and UHMEP just before sending the message and before processing it. This implementation guide can be used as a JSON validator on both end of the communication. All information about how to use it is available on the link:

https://build.fhir.org/ig/hl7-be/referral/branches/nursing-pilot/index.html

6.3.1 Capability Statement

As a FHIR compliant server, UHMEP makes available a statement of the accessible features.

This method is accessible to everyone, without any token from eHealth. The endpoint is the following:

GET /metadata

This returns a CapabilityStatement resource detailing the supported resources and operations.

6.3.2 A prescription in FHIR resources

A Belgian referral prescription is represented by the BeReferralServiceRequest resource.. A BeReferralTask is always made available on creation of a prescription. It has the same ID as the one of the BeReferralServiceRequest and is linked to the BeReferralServiceRequest through its ***focus*** field and can be consulted with the endpoint:

GET /Task/{ID}

On prescription creation, BePractitionerRole and BePractitioner resources are also made available for consultation as described in section 6.3.3.

6.3.2.1 Prescription statuses

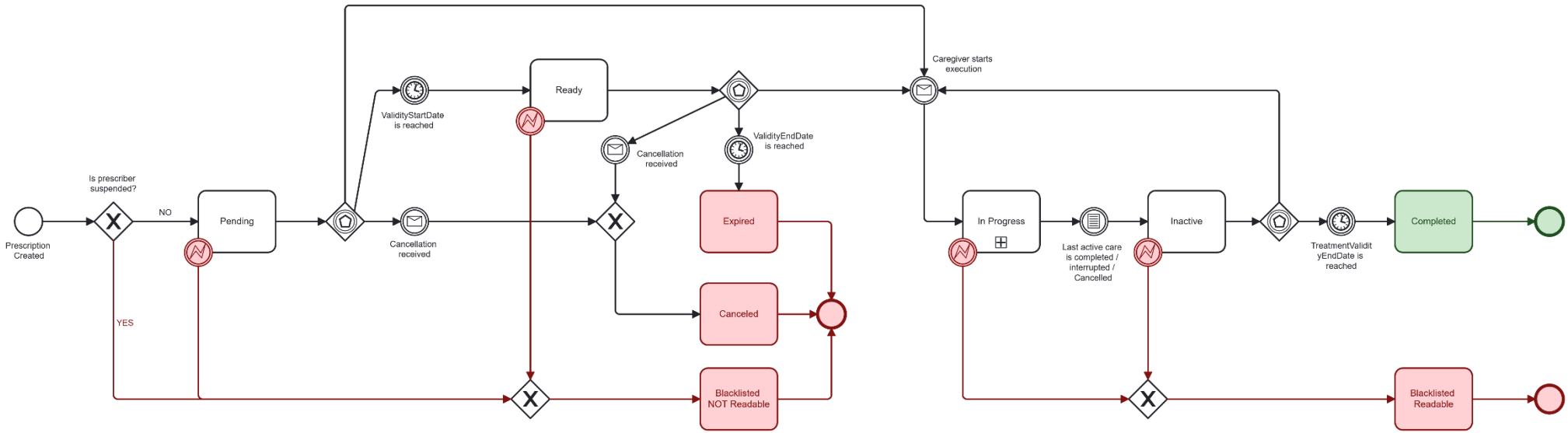
To follow the lifecycle of a prescription, every business status has a mapping with FHIR available statuses through many resources (BeReferralServiceRequest + BeReferralTask).

The current mapping between business and FHIR statuses is:

Business Status	*BeReferralServiceRequest.status*	*BeReferralServiceRequest.statusReason*	*BeReferralTask.status*
Draft	“draft”	/	“draft”
Blacklisted	“entered-in-error”	/	“entered-in-error”
Pending	“active”	/	“draft”
Ready	“active”	/	“ready”
Canceled	“revoked”	“cancelingReason”	“cancelled”
Expired	“revoked”	“expired”	“cancelled”
In Progress	“active”	/	“in-progress”
Inactive	“active”	/	“on-hold”
Completed	“completed”	/	“in-progress”/”completed”

In the cookbook, business statuses will be used when talking about statuses to be more aligned with the requirements document and to lighten the explanation by avoiding expressing the value of each resource.

6.3.2.2 Prescription lifecycle



6.3.2.3 Execution of caregivers on prescriptions

The Belgian referral prescription is mainly stored in the BeReferralServiceRequest. Once created by the prescriber, clinical data in this resource are fixed and should not be changed afterward. This prescription will start its "prescription status lifecycle" like described in the business requirements release.

Another resource is used to handle caregiver interactions with the prescription, namely the BePerformerTask. This resource is linked to the BeReferralServiceRequest through a reference in its ***focus*** field and a reference to the BeReferralTask in its ***partOf*** reference.

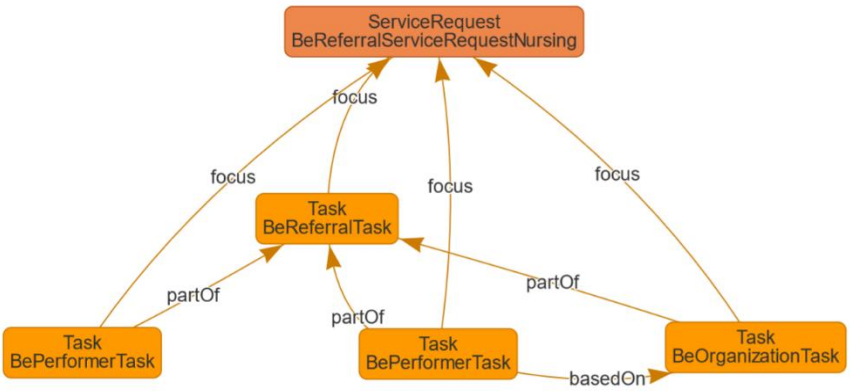


Figure 2: Links between Task resources to the BeReferralServiceRequest

Similarly, the BeOrganizationTask resource is used to store an assignation of an organization on a prescription. If a caregiver is working for an organization, his BePerformerTask should be linked to the BeOrganizationTask through the ***focus*** field. The Figure 2 above shows links between BePerformerTasks, BeReferralTask, BeOrganizationTask and BeReferralServiceRequest when there are two executions on a prescription, when linked to an assignation to an organization.

All information about operations and consultation that can be done on this resource is available in section 8.

The BePerformerTask resource can be created in three cases:

- At the assignation of a caregiver on a prescription by the Patient or by another Caregiver
- When the caregiver starts his execution, and the resource wasn't already created
- When the caregiver completes his execution, without prior assignation nor start (this is a shortcut)

After this, all operations are done using the PATCH operation of FHIR.

The table below is the mapping between business statuses and FHIR statuses

Business Status	BePerformerTask.status
Ready	“ready”
In Progress	“in-progress”
Canceled	“cancelled”
Interrupted	“on-hold”
Completed	“completed”

The Figure 3 below shows the status flow of a BePerformerTask. Once the BePerformerTask ***status*** is “on-hold” or “completed”, the resource cannot be updated anymore.

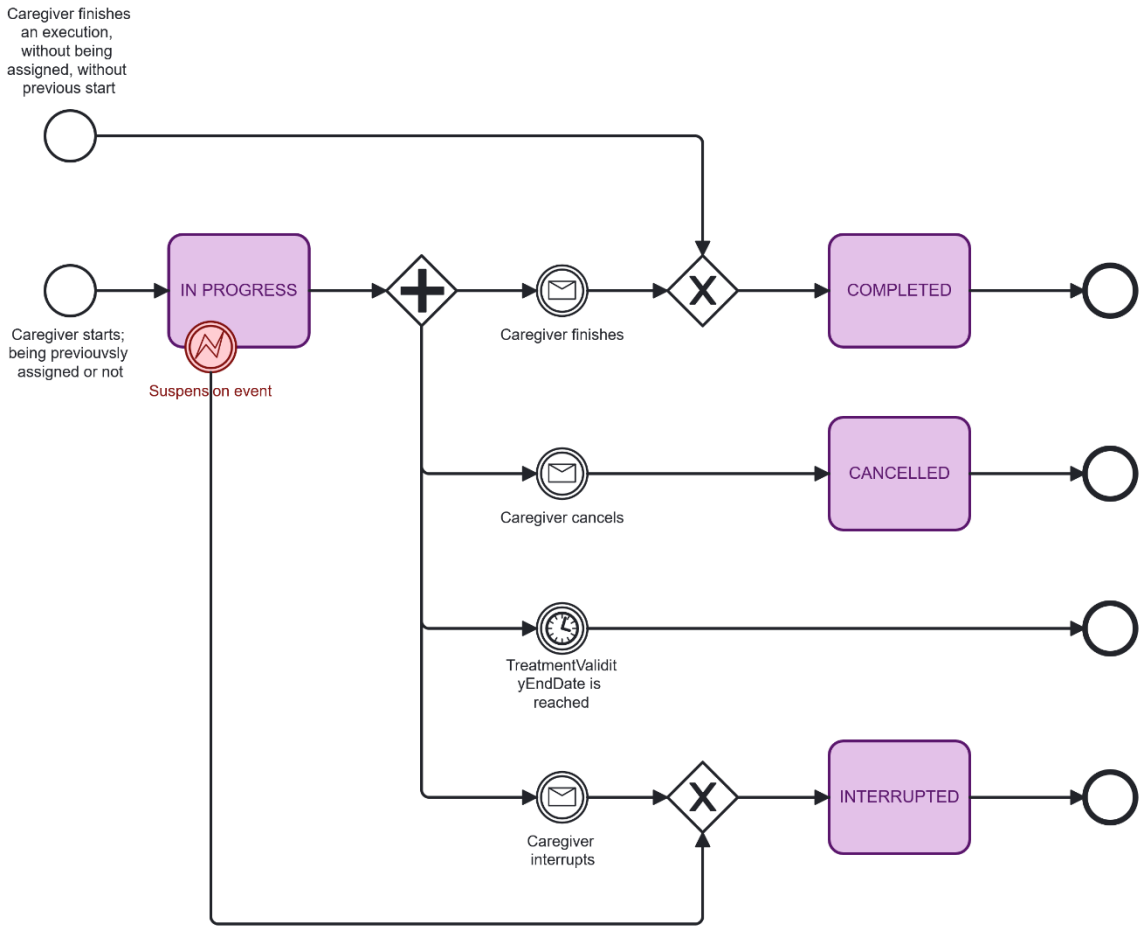


Figure 3: Status diagram of BePerformerTask

6.3.3 References to healthcare professionals

As mentioned above, UHMEP uses FHIR as the transport layer for all communications. In resources like the BeReferralServiceRequest and BePerformerTask, references to healthcare professionals are done with references to BePractitionerRole FHIR resource.

IDs of BePractitionerRole in UHMEP are the concatenation between the SSIN and the discipline of the healthcare professional. For example, for a “NURSE” with “82042605839” as SSIN, the ID of the BePractitionerRole in UHMEP for this person would be : “82042605839-NURSE”.

Disciplines are defined by eHealth/INAMI and available through CoBRHA. Disciplines should be in UPPERCASE and all spaces replaced by underscores ("_").

For all resources, references can be put directly like "PractitionerRole/82042605839-NURSE". In the BePerformerTask for example, the reference of the ***owner*** field will have the format like below:

```
"owner": {
    "reference": "PractitionerRole/82042605839-NURSE"
}
```

Snippet 1: Owner reference example in a BePerformerTask

The BePractitionerRole is created by UHMEP and can be consulted using the usual FHIR resource consultation endpoint format:

GET /PractitionerRole/82042605839-NURSE

This resource itself does a reference to a ***practitioner*** that is a reference to a BePractitioner FHIR resource. IDs of BePractitioner are only the SSIN of the healthcare professional.

An example is available below:

```
"practitioner": {
    "reference": "Practitioner/82042605839"
}
```

Snippet 2: Practitioner reference example in a BePractitionerRole

Similarly, the BePractitioner resource is created by UHMEP and can be consulted using the usual FHIR resource consultation endpoint format:

GET /Practitioner/82042605839

Those endpoints cannot be called by a patient. Only a healthcare professional can call those endpoints.

6.3.4 References to organizations

BeOrganizationTask were introduced in section 6.3.2.3, and are used to store assignation of organizations on prescriptions. References to an organization should be a reference to a BeOrganization FHIR resource.

IDs of BeOrganization in UHMEP are the concatenation between the NIHII and the organization type of the organization. For example, for an organization with a type “940” and “94754895” as NIHII, the ID of the BeOrganization UHMEP for this organization would be : “NIHII-94754895-940”. “NIHII” is not case sensitive.

Organization types are defined by eHealth/INAMI and available through CoBRHA.

For all resources, references can be put directly like "Organization/NIHII-94754895-940". In the BeOrganizationTask for example, the reference of the ***owner*** field will have the format like below:

```
"owner": {
    "reference": " Organization/NIHII-94754895-940"
}
```

Snippet 3: Owner reference example in a BePerformerTask

The BeOrganization is created by UHMEP and can be consulted using the usual FHIR resource consultation endpoint format:

GET /Organization/NIHII-94754895-940

6.3.5 Pseudonymization of SSIN

To protect patient privacy, UHMEP requires SSIN pseudonymization.

All direct pseudonyms (representing an underlying identifier like an SSIN or an encryption key that has been processed by the pseudonymization service) **must** follow this structure:

urn:be:fgov:pseudo:v2:{SEC1}:{transitInfo}

- urn:be:fgov:pseudo:v2: Standard prefix indicating a version 2 direct pseudonym.
- {SEC1}: The first part of the pseudonym service's output.
- {transitInfo}: The second part, a JWE compact string. This JWE contains metadata critical for the pseudonymization service and the recipient

Please refer to the eHealth Pseudonymisation REST Cookbook, referenced in section 2.1.

6.3.5.1 FHIR example

Following eHealth FHIR recommendations for pseudonym handling, here is an example of how patient identifiers should be used in FHIR resources in UHMEP:

```
"subject": {
  "identifier": {
    "use": "official",
    "system":
"https://www.ehealth.fgov.be/standards/fhir/core/NamingSystem/ssin",
    "value": "urn:be:fgov:pseudo:v2:{SEC1}:{transitInfo}",
    "_value": {
      "extension": [
        {
          "url":
"https://www.ehealth.fgov.be/standards/fhir/infsec/StructureDefinition/be-ext-
pseudonymization",
          "extension": [
            {
              "url": "marker",
              "valueBoolean": true
            },
            {
              "url": "format",
              "valueCode": "direct"
            },
            {
              "url": "version",
              "valuePositiveInt": 2
            }
          ]
        }
      ]
    }
  }
}
```

6.3.6 Pseudonymization impacts on free text fields

In the free text fields, it is possible that the prescriber adds some sensitive data about the patient.

To avoid any exposure of sensitive data, all free text fields are encrypted and the key is pseudonymized. In FHIR, it means that each field which can contain a text must be attached with this FHIR extension "be-ext-pseudonymization". Fields concerned:

- BeCodedAnnotation.text
- CodeableConcept.text

Following eHealth Guidance, the key used for the encryption is provided in a pseudonymized way in the FHIR extension "be-ext-pseudonymization" which is contained in the FHIR extension "be-ext-key-pseudonymization" which is in the field "meta" of the ServiceRequest.

For each free text field, the identifier of the key (the "kid") must be the same as the one provided in the "meta" field. **All systems should use and reuse the key provided in the "meta" field of a FHIR resource**. Indeed, one key can encrypt and decrypt multiple free text in one FHIR resource. If no key has been already attached, or at the creation of a resource, it is up to the system to generate one following eHealth guidance (can be found in the Annex 9 of the Pseudonymisation REST Cookbook).

If multiple keys are present, or no key is linked to a free text, UHMEP will reject the request.

References to encrypted free text content **must** follow this structure:

urn:be:fgov:pseudo-encrypted:v1:{KID}:{JWE}

- urn:be:fgov:pseudo-encrypted:v1: Standard prefix indicating a version 1 encrypted payload reference.
- {KID}: The **Key Identifier** (a UUID) of the symmetric encryption key that was used to encrypt the actual free text. This KID allows systems to locate the pseudonymized encryption key.
- {JWE}: The compact JWE string representing the actual encrypted free text content.

6.3.6.1 FHIR Examples

6.3.6.1.1 Pseudonymized Encryption Key

```
{
  "resourceType": "ServiceRequest",
  "meta": {
    "extension": [
      {
        "id": "370f2865-03ff-4c7b-affd-897a4aae187e", // The KID for this key
        "url": "https://www.ehealth.fgov.be/standards/fhir/infsec/StructureDefinition/be-ext-
key-pseudonymization",
        "extension": [
          {
            "url": "key",
            "valueString": "urn:be:fgov:pseudo:v2:{SEC1}:{transitInfo} ",
            "_valueString": {
              "extension": [
                {
                  "url":
"https://www.ehealth.fgov.be/standards/fhir/infsec/StructureDefinition/be-ext-
pseudonymization",
                  "extension": [
                    {
                      "url": "marker",
                      "valueBoolean": true
                    },
                    {
                      "url": "format",
                      "valueCode": "direct"
                    },
                    {
                      "url": "version",
                      "valuePositiveInt": 2
                    }
                  ]
                }
              ]
            }
          }
        ]
      }
    ]
  }
  // ... other resource fields, including encrypted text referencing this KID ...
}
```

6.3.6.1.2 Encrypted free text

```
"text": "urn:be:fgov:pseudo-encrypted:v1:370f2865-03ff-4c7b-affd-897a4aae187e:{JWE}", // The same KID for the key, alongside the encrypted free text in JWE format
"_text": {
  "extension": [
    {
      "url": "https://www.ehealth.fgov.be/standards/fhir/infsec/StructureDefinition/be-ext-pseudonymization",
      "extension": [
        {
          "url": "marker",
          "valueBoolean": true
        },
        {
          "url": "format",
          "valueCode": "encrypted"
        },
        {
          "url": "version",
          "valuePositiveInt": 1
        }
      ]
    }
  ]
}
```

7 Technical requirements

7.1 Tracing

To use this service, the request MUST contain the following two http header values (see RFC https://datatracker.ietf.org/doc/html/rfc7231#section-5.5.3):

1. **User-Agent**: information identifying the software product and underlying technical stack/platform.

    a. Pattern: {company}/{package-name}/{version} {platform-company}/{platform-package-name}/{platform-package-version}

    b. Regular expression for each subset (separated by a space) of the pattern: [[a-zA-Z0-9-\/]*\/[0-9a-zA-Z-_.]*

    c. Examples:
    User-Agent: MyCompany/myProduct/62.310.4 eHealth/Technical/3.19.0
    User-Agent: Topaz-XXXX/123.23.X Taktik/freeconnector/XXXXX.XXX

2. **From:** email-address that can be used for emergency contact in case of an operational problem.
    Examples:
    From: info@mycompany.be

7.2 API documentation

The last version of REST interface described with a JSON / Swagger API is available on the eHealth API Portal:

Environment	Endpoint
Acceptance	https://portal-acpt.api.ehealth.fgov.be
Production	https://portal.api.ehealth.fgov.be

Endpoint to call UHMEP is described in the API documentation on the eHealth API Portal.

Different operations are available and textually described in section 8.

8 Description of the operations

8.1 Common validations

For all incoming requests, UHMEP applies a suite of common validations if the request is validated by the implementation guide described in section 6.3.

8.1.1 Validation of dates

In the payload, different types of date are used.

- Standard Dates: the format must be YYYY-MM-DD.
- Recorded date (***authoredOn*** field): the format must be YYYY-MM-DDThh:mm:ss+zz:zz (for example: `2023-05-12T13:37:42+02:00`) .
    - It is forbidden to create a resource with a timestamp in the future.

8.1.2 Validation of the template

UHMEP is using FHIR resources to exchange data between the applications. The resources follow INAMI's templates of referral prescriptions. The template must be known from UHMEP. If it is not the case, the user will receive an error.

FHIR profiles have been designed to match the business requirements described by INAMI in their requirements documents.
Each profile addresses a specific discipline or a specific use case.

Currently, the following FHIR profiles have been designed:

- Nursing Prescription: BeReferralServiceRequestNursing
- Annex 81 (Proposal): BeAnnex81
- Diagnostic Imaging Prescription: BeReferralServiceRequestDiagnosticImaging

When integrators submit a creation request, the FHIR resource they send must respect to the constraints described in those profiles.
Integrators are responsible for using the right profile according to the prescription they aim to create.

To help integrators and UHMEP system to identify the type of prescription represented by a FHIR ServiceRequest, we introduced a technical identifier in the resource under the field "category".
This technical identifier must always be provided at the creation of the ServiceRequest.

The list of technical identifier contains the following:

Identifier	Prescription Type
hygienic-care	Hygienic care
bloodletting	Bloodletting
chronic-dialysis	Chronic home dialysis
diabetic-education-within-care-trajectory	Diabetes education within a care trajectory
diabetic-education-within-start-trajectory	Diabetes education within a start trajectory

diabetic-education-without-care-trajectory	Diabetes education for patients without a trajectory
diabetic-education-convention-center	Diabetes education within a convention centre
glycemic-test	Glycemic control with glucometer
medication-prefill-preparation-appendix-81	Preparing medication (Appendix 81)
medication-prefill-preparation-not-refunded	Preparation of medicines (Not refunded)
parameters	Measurement of parameters
sampling	Sample collection
generic-nursing	Generic nursing care
diagnostic-imaging-appendix-82	Diagnostic Imaging (Appendix 82)

8.1.2.1 For nursing care

8.1.2.1.1 Prescriptions

Prescription Type
Hygienic care
Bloodletting
Chronic home dialysis
Diabetes education within a care trajectory
Diabetes education within a start trajectory
Diabetes education for patients without a trajectory
Diabetes education within a convention centre
Glycemic control with glucometer
Preparation of medicines (Not refunded)
Measurement of parameters
Sample collection
Generic nursing care

The “Generic nursing care” prescription serves as a placeholder for the remaining templates that are yet to be implemented on UHMEP.

To create one of these prescriptions, integrators must use the profile BeReferralServiceRequestNursing.

For example, “Diabetes education via convention center” prescription is a BeReferralServiceRequestNursing with:

- ***code:***
  - *code* = “385805005”
  - *system* = “http://snomed.info/sct”
- ***orderDetail***
  - *code* = “211141000172109”
  - *system* = “http://snomed.info/sct””

- ***category:***
  - *category[discipline]:*
    - *code* = "9632001"
    - system = "http://snomed.info/sct"
  - *category[prescriptionType]:*
    - *code* ="diabetic-education-convention-center"
    - *system* = "https://www.ehealth.fgov.be/standards/fhir/terminology/CodeSystem/be-cs-prescription-type"

The **category[prescriptionType]** defines the template of the BeReferralServiceRequestNursing resource.

8.1.2.1.2 Annex 81 (Proposal)

Prescription Type
Preparing medication (Appendix 81)

To create one of this annex 81, integrators must use the profile BeAnnex81.

For example, "Diabetes education via convention center" prescription is a BeAnnex81 with:

- ***code:***
  - *code* = "385796006"
  - *system* = "http://snomed.info/sct"
- ***category:***
  - *category[discipline]:*
    - *code* = "9632001"
    - system = "http://snomed.info/sct"
  - *category[annex81TechnicalType]:*
    - *code* ="medication-prefill-preparation-appendix-81"
    - *system* = "https://www.ehealth.fgov.be/standards/fhir/terminology/CodeSystem/be-cs-prescription-type"

8.1.2.2 For radiology

Prescription Type
diagnostic-imaging-appendix-82

To create one of this annex 82, integrators must use the profile BeReferralServiceRequestDiagnosticImaging.

For example, "Diabetes education via convention center" prescription is a BeReferralServiceRequestDiagnosticImaging with:

- ***code:***
  - *code* = "116152004" (example)
  - *system* = "http://snomed.info/sct"
- ***category:***
  - *category[discipline]:*
    - *code* = "363679005"
    - system = "http://snomed.info/sct"
  - *category[*prescriptionType*]:*
    - *code* ="diagnostic-imaging-appendix-82"
    - *system* = "https://www.ehealth.fgov.be/standards/fhir/terminology/CodeSystem/be-cs-prescription-type"

8.1.3 Validation of the prescriber and caregiver information

To optimize performance and minimize unnecessary database calls, UHMEP first validates the SSIN format using a checksum algorithm.
The API then verifies the existence of the healthcare professional in the eHealth authentic source CoBRHA. The SSIN and discipline in the payload (or token) must match CoBRHA.

8.1.4 Validation of the organization

UHMEP verifies the healthcare organization in CoBRHA. The NIHII number and organization type provided must follow INAMI's requirements and exist in CoBRHA.

In addition, UHMEP verifies if a given organization is part of a CoT.

8.1.5 Validation on the patient

UHMEP verifies the patient's existence in its internal database. If not found, it consults the National Register via the eHealth ConsultRN service using the pseudonymized SSIN. If the patient is unknown, the prescription cannot be created.

UHMEP handles SSIN replacement. Consultation of a list of prescription (section 8.5.2) based on the replaced or the new SSIN will both return all prescription created for both old and new SSIN for the same patient.

Pseudonyms in transit generated for UHMEP are strictly exclusive to the specific patient and cannot be identified by other citizens.

8.1.6 Authentication

Each user using UHMEP must be authenticated to be allowed to use UHMEP. UHMEP will verify the validity of the access token. UHMEP will also use the other information contained in the token to accept or reject the request and applies the Access Matrix (see Section 6.2.3). The header should be:

```
Authorization: Bearer <exchanged_ehealth_token>
```

8.1.7 Visible prescriptions

8.1.7.1 Depending on the template

When a prescriber/caregiver can access an operation because he has the correct role, only the prescriptions that he is allowed to see are returned. To know if he is allowed to see the prescription, a check is done by looking the consultation right in the access matrix like it is described in section 6.2.3.

8.1.7.2 Blacklisted prescriptions

When a prescriber/caregiver can access an operation because he has the correct role, some prescriptions in the status “Blacklisted” will not be returned.

A prescription can be blacklisted at the creation because the prescriber was suspended (see the suspension explanation in section 6.2.4) but it can also be blacklisted after its creation due to a retroactive script executed by UHMEP. Only prescriptions created in the suspension period of the prescriber will be blacklisted.

This script is launched when the information of the prescriber suspension is received only some days after the creation.

Depending on the actual status of the prescription, the blacklisting script will have different behaviors:

- If the prescription was in one of the business statuses “Draft”, “Ready”, “Cancelled”, “Expired”:
  - The prescription is blacklisted.
  - The prescription will not be returned (in case of a specific consultation, an error will be returned).
- If the prescription was in the business status “In Progress”:
  - The prescription is blacklisted.
  - The status of all executions in “in-progress” is changed to “interrupted”.
  - The prescription will be returned.
- If the prescription was in one of the business statuses “Inactive”, “Completed”:
  - The prescription is blacklisted.
  - The prescription will be returned.

To summarize, when a prescription is in the status “Blacklisted”, it will be returned if there is one or more executions. If there is nothing or only assignations on it, the prescription will not be returned.

8.1.7.3 Cancelled prescriptions

When a referral prescription reaches the "Cancelled" (revoked) status, its visibility is strictly restricted to ensure confidentiality:

- **Patients:** Can see the cancelled prescription (if they are the subject).
- **Prescribers:** Can see the cancelled prescription **only** if they are the original creator (requester) of the prescription. Other prescribers cannot see it.
- **Caregivers:** Cannot see cancelled prescriptions.

8.2 Operations on prescriptions

8.2.1 Creation of a prescription

This method creates a new referral prescription in the UHMEP database. Upon successful creation, the system will asynchronously compute an eHealth timestamp (digital signature) based on the payload to guarantee its integrity and authenticity.

8.2.1.1 Allowed roles

Prescriber, Caregiver (only for radiology prescriptions), Organization

8.2.1.2 Endpoint

POST /ServiceRequest

8.2.1.3 Implemented rules

- **Token Match:** The SSIN and Discipline of the healthcare professional in the payload must exactly match the CoBRHA database and the active eHealth token.
- **Suspension Check:** The prescriber must not be suspended. *(Note: If they are suspended, the prescription WILL be created, but forced into the "Blacklisted" status and an error will be returned).*
- **Privacy Checks:** Informed Consent, Therapeutic Relations, and Therapeutic Exclusions are validated (See Section 6.2.5).
- **Intent:** The creation of a resource with an intent other than "order" (for prescriptions) or "proposal" (for Annex 81 proposals) is forbidden. This method expects "order".
- **Template Identification:** The prescription template must be identified using the ServiceRequest.category field.
- **Mandatory Medical Reason (Frequency Rule):** If the prescription occurrenceTiming.repeat.frequency is **≥ 2 per day**, a medical reason/diagnosis is **mandatory**.
    - *Exceptions:* This rule is ignored for templates where the medical reason field does not exist: *Medication prefill preparation*, *Diabetes education without care path and Diabetes education via convention center*
- **Extensions (basedOn):** If this prescription is an extension of a previous one, the basedOn field must be used to reference the original ServiceRequest ID. Both prescriptions must belong to the exact same patient and use the exact same template. *(Note: Annex 81 proposals/approvals cannot be extended or duplicated).*

- **Validity Dates:**
  - ***validity.start*** can be a maximum of 5 days before the recorded date (***authoredOn***).
  - If ***validity.start*** is empty, UHMEP will auto-fill it with the ***authoredOn*** date (converted to YYYY-MM-DD).
  - The maximum allowed ***validity.end*** date depends strictly on the template type:
    - **Standard Nursing Templates:** ***validity.start*** + 2 years - 1 day. *(e.g., Hygienic Care, Parameters, Sampling).*
    - **1-Year Templates:** ***validity.start*** + 1 year exact. *(e.g., Annex 81, Diabetic Education via Convention Center, Medication Prefill).*
    - **End-of-Year Templates:** December 31st of the validity.start year. *(e.g., Diabetic Education with Care Path / Model of Care).*
  - Logic: ***validity.start*** <= ***validity.end*** <= ***latest***.
  - Logic: ***validity.end*** AND ***latest*** > ***authoredOn***.

8.2.1.4 Request

The request must contain the information of the prescription in the body using the BeReferralServiceRequest resource. All free texts should be encrypted, following the description in section 6.3.6.

8.2.1.5 Response

On successful creation of the BeReferralServiceRequest resource, the server will respond with a HTTP 201. The ID of this newly created resource is accessible in the *Location* header. This ID can be used to further interactions with this resource. Additionally, the generated ID is now returned directly inside the JSON response payload.

A short code is also generated by the backend and will be returned at the consultation of the prescription.

8.2.2 Cancel a prescription

This method allows the end-user to cancel a referral prescription via his user interface in the UHMEP application. It means that the end-user would be able to continue to consult it but the prescription will no longer be executable.

8.2.2.1 Allowed roles

Prescriber, Patient and Organization.

8.2.2.2 Endpoint

PATCH /ServiceRequest/{prescriptionID}

8.2.2.3 Implemented rules

The cancellation of a specific prescription is possible only if all the following points are respected:

- The prescription is in one of the business statuses:
  - "Draft"
  - "Pending"
  - "Ready"
- **Execution Blocker:** A prescription **cannot** be canceled if any execution (BePerformerTask) has advanced past the "ready" status. Even if a caregiver's treatment execution was subsequently "cancelled" or "on-hold", the prescription itself can no longer be canceled by the prescriber/patient.
- (Prescriber) He is not suspended.
- (Prescriber) Must be the original creator/requester of the prescription.
- (Patient) The prescription must belong to the connected patient.
- Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations, please refer to section 6.2.5.

8.2.2.4 Request

The operation is done through a PATCH on the BeReferralServiceRequest resource. It should only replace the ***status*** of this resource to the one "revoked". An example is available in Snippet 4:

```
{
    "resourceType": "Parameters",
    "parameter": [
        {
            "name": "operation",
            "part": [
                {
                    "name": "type",
                    "valueCode": "replace"
                },
                {
                    "name": "path",
                    "valueString": "ServiceRequest.status"
                },
                {
                    "name": "value",
                    "valueString": "revoked"
                }
            ]
        }
    ]
}
```

Snippet 4: PATCH example on a BeReferralServiceRequest to cancel a prescription

8.2.2.5 Response

On successful cancellation of the prescription, the server returns HTTP 200.

8.3 Assignation of a caregiver on a prescription

8.3.1 Add a caregiver

Once a prescription has been created, the system allows users to assign a specific caregiver on it. Giving them an ease of access to this prescription and some access rights.

8.3.1.1 Allowed roles

Patient, Caregiver and Organization.

8.3.1.2 Endpoint

POST /Task

8.3.1.3 Implemented rules

The assignation of a caregiver on an existing prescription is possible only if all following rules are respected:

- (Patient/Caregiver) The prescription referenced in the ***focus*** field is in one of the business statuses:
  - "Draft"
  - "Pending"
  - "Ready"
  - "In progress"
  - "Inactive"
  - "Completed"
- (Patient/Caregiver) The ***status*** of the newly created BePerformerTask in the request is "ready".
- (Patient/Caregiver) No ***executionPeriod*** is provided.
- (Patient/Caregiver) No other BePerformerTask in ***status*** ["ready", "in-progress", "completed"] is already linked with ***focus*** to the same prescription for the same ***owner***.
- (Patient/Caregiver) The caregiver to assign is not currently suspended.
- (Patient/Caregiver) The caregiver to assign exists in CoBRHA database.
- (Patient/Caregiver) The discipline of the caregiver to assign is allowed to consult following the Access Matrix table for the template of the prescription.
- (Patient) The connected patient is the same as the one of the prescriptions.
- (Caregiver) The healthcare professional connected is not suspended.
- Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations, please refer to section 6.2.5.

8.3.1.4 Request

The assignation is done through the creation of a BePerformerTask resource.

For more information about how BePerformerTask are handled in UHMEP, see section 6.3.2.3.

Particularities coming from UHMEP:

Field	Particularity
focus	This is a reference to the prescription, the BeReferralServiceRequest. (ex: ServiceRequest/{referenceID})
partOf	This is a reference to a BeReferralTask, that currently has the same ID as the prescription. (ex: Task/{referenceID})
intent	UHMEP only allows “order”

8.3.1.5 Response

On successful creation of the BePerformerTask resource, the server will respond with a HTTP 201. The ID of this newly created resource is accessible in the *Location* header. This ID can be used for further interactions with this resource. Additionally, the generated ID is now returned directly inside the JSON response payload.

8.3.2 Remove a caregiver

The inverse operation is also possible to remove an assigned caregiver from a prescription.

8.3.2.1 Allowed roles

Patient, Caregiver and Organization

8.3.2.2 Endpoint

DELETE /Task/{executionID}

{executionID} being the ID of the BePerformerTask to be removed from a BeReferralServiceRequest.

8.3.2.3 Implemented rules

The removal of an assigned caregiver on an existing prescription is possible only if all following rules are respected:

- The prescription referenced in the ***focus*** field is in one of the business statuses:
  - "Draft"
  - "Pending"
  - "Ready"
  - "In Progress"
  - "Inactive"
- The execution of the caregiver (BePerformerTask) is in ***status*** "ready".
- The user connected must not be suspended.
- The patient connected must own the prescription.
- No Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations are performed during the removal of an assignation, please refer to section 6.2.5.

8.3.2.4 Request

In addition of the authorization and header described above, this request does not need a body.

8.3.2.5 Response

On successful deletion of the resource, the server sends back a HTTP 204. After this operation, the resource is definitively deleted and cannot be used anymore.

8.4 Execution of a prescription

Caregivers cannot modify the BeReferralServiceRequest (the prescription itself) directly. All execution actions (Start, Finish, Interrupt, Revoke) are performed by creating or patching their assigned BePerformerTask.

8.4.1 Start an execution

The start operation allows the caregiver to indicate to UHMEP that he works or has started his work on the prescription.
His execution will be in the ***status*** “in-progress”.

There are two ways in UHMEP to start an execution:

- By creating a BePerformerTask
- By updating a BePerformerTask thanks to a PATCH method

For more information about BePerformerTasks handling by UHMEP, see section 6.3.2.3.

The operation of “starting an execution” only adds a start date on the execution. This start date can be in the past, to allow caregivers to do their administrative work afterward.

8.4.1.1 Start an execution through a creation of resource

If the caregiver was not previously assigned to the prescription, they can start execution immediately by creating a Task directly in the "in-progress" status.

8.4.1.1.1 Allowed Role

Caregiver and Organization

8.4.1.1.2 Endpoint

POST /Task

8.4.1.1.3 Implemented rules

The creation of an execution task is possible only if all the following points are respected:

- The caregiver is/was not suspended on the date provided in the ***executionPeriod.start***.
- The prescription referenced in the ***focus*** field is in one of the business statuses:
  - “Pending”
  - “Ready”
  - “In progress”
  - “Inactive”
  - “Expired” (Only if the ***executionPeriod.start*** is before the ***extension.validity.end*** date of the BeReferralServiceRequest)
- No other execution task in “ready” or "in-progress” exists for this caregiver (execution with the same reference in the ***focus*** field).
- The task ***status*** is "in-progress".
- Only the ***executionPeriod.start*** date is provided for the execution dates.

- ***executionPeriod.start*** date is not in the future but can be in the past.
  If it is in the past:
  - [***BeReferralServiceRequest.authoredOn <= BeReferralServiceRequest.extension.validity.start***] ***executionPeriod.start*** cannot be before the ***authoredOn*** date of the BeReferralServiceRequest.
  - [***BeReferralServiceRequest.extension.validity.start < BeReferralServiceRequest.authoredOn***] ***executionPeriod.start*** cannot be before the ***extension.validity.start*** date of the BeReferralServiceRequest.
- Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations, please refer to section 6.2.5.
- The information contained in the ***owner*** field matches with the information of the connected caregiver in the token.
- Upon starting the first execution, the backend will automatically calculate and lock in the TreatmentValidityEndDate for the prescription based on the template rules (e.g., Start Date + 5 years, + 1 year, or End of Year)

8.4.1.1.4 Request

In addition of the authorization and header described above, this request needs a body containing the information of the task to create.

The execution task is represented in FHIR by the Belgian profile of the Task resource called BePerformerTask.

Particularity coming from UHMEP:

Field	Particularity
Intent	UHMEP will only accept “order”

8.4.1.1.5 Response

On successful creation of the BePerformerTask resource, the server will respond with a HTTP 201. The ID of this newly created resource is accessible in the *Location* header. This ID can be used for further interactions with this resource.

8.4.1.2 Start an execution through a PATCH

If the caregiver is already assigned on the prescription, this method must be used to update the existing BePerformerTask (see section 6.3.2.3).

8.4.1.2.1 Allowed roles

Caregiver and Organization.

8.4.1.2.2 Endpoint

PATCH /Task/{executionID}

{executionID} being the ID of the BePerformerTask to be updated from a BeReferralServiceRequest.

8.4.1.2.3 Implemented rules

The update of an execution task is possible only if all the following points are respected:

- The Task must have previously the status “ready” or “cancelled”.
- All rules as in section 8.4.1.1.3 are respected.

8.4.1.2.4 Request

The operation is done through a PATCH on the BePerformerTask FHIR resource. It should only add the start execution date (YYYY-MM-DD) on this resource. An example is available below:

```
{
    "resourceType": "Parameters",
    "parameter": [
        {
            "name": "operation",
            "part": [
                {
                    "name": "type",
                    "valueCode": "add"
                },
                {
                    "name": "path",
                    "valueString": "Task.executionPeriod"
                },
                {
                    "name": "name",
                    "valueString": "start"
                },
                {
                    "name": "value",
                    "valueDateTime": "{{StartExecutionDate}}"
                }
            ]
        }
    ]
}
```

Snippet 5: PATCH example on a BePerformerTask to start an execution

8.4.1.2.5 Response

On successful update of the execution, the server sends back a HTTP 200. The response does not contain a body.

8.4.2 Finish an execution

The finish operation allows the caregiver to indicate to UHMEP that he has finished his work on the prescription.
His execution will be in the ***status*** “completed”.

There are two ways in UHMEP to finish an execution:

- By creating a task
- By updating a task thanks to a PATCH method

8.4.2.1 Finish an execution through a creation of resource

If the caregiver is not assigned on the prescription, this method must be used to create the execution directly in the status “completed”.

8.4.2.1.1 Allowed roles

Caregiver and Organization.

8.4.2.1.2 Endpoint

POST /Task

8.4.2.1.3 Implemented rules

The creation of an execution task is possible only if all the following points are respected:

- The caregiver is not suspended.
- The prescription referenced in the ***focus*** field is in one of the business statuses:
  - “Ready”
  - “In Progress”
  - “Inactive”
  - “Expired”
- No other execution task in “ready”, "in-progress", “completed” and “cancelled” exists for this caregiver (execution with the same reference in the ***focus*** field).
- The BePerformerTask ***status*** is "completed" and the ***executionPeriod.start*** and ***executionPeriod.end*** dates are provided for the execution dates.
- Execution dates are not in the future.
- ***executionPeriod.start*** <= ***executionPeriod.end***.
- ***executionPeriod.end <= BeReferralServiceRequest.extension.latest.***
- ***executionPeriod.start*** date is not in the future but can be in the past.
  If it is in the past:
  - [***BeReferralServiceRequest.authoredOn <= BeReferralServiceRequest.extension.validity.start***] ***executionPeriod.start*** cannot be before the ***authoredOn*** date of the BeReferralServiceRequest.
  - [***BeReferralServiceRequest.extension.validity.start < BeReferralServiceRequest.authoredOn***] ***executionPeriod.start*** cannot be before the ***extension.validity.start*** date of the BeReferralServiceRequest.

- The information contained in the ***owner*** field matches with the information of the connected caregiver in the token.
- Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations, please refer to section 6.2.5.

8.4.2.1.4 Request

In addition of the authorization and header described above, this request needs a body containing the information of the task to create.

The execution task is represented in FHIR by the Belgian profile of the task resource called BePerformerTask.

Particularity coming from UHMEP:

Field	Particularity
intent	UHMEP will only accept the value “order"

8.4.2.1.5 Response

On successful creation of the BePerformerTask resource, the server will respond with a HTTP 201. The ID of this newly created resource is accessible in the *Location* header. This ID can be used for further interactions with this resource.

8.4.2.2 Finish an execution through a PATCH

If the caregiver has already started an execution on the prescription, this method must be used to update the existing execution.

8.4.2.2.1 Allowed roles

Caregiver and Organization.

8.4.2.2.2 Endpoint

PATCH /Task/{executionID}

{executionID} being the ID of the BePerformerTask to be updated from a BeReferralServiceRequest.

8.4.2.2.3 Implemented rules

The update of an execution task is possible only if all the following points are respected:

- The caregiver is not suspended.
- The prescription referenced in the ***focus*** field is in one of the business statuses:
  - "In Progress"
- The BePerformerTask is in the ***status*** "in-progress".
- Only the ***executionPeriod.end*** date is provided in the PATCH request for the execution dates.
- ***executionPeriod.end*** date is not in the future.
- ***executionPeriod.start*** <= ***executionPeriod.end***.
- ***executionPeriod.end <= BeReferralServiceRequest.extension.latest***
- The information contained in the ***owner*** field must match with the information of the connected caregiver in the token.
- Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations, please refer to section 6.2.5.

8.4.2.2.4 Request

The operation is done through a PATCH on the BePerformerTask FHIR resource. It should only add the end execution date (YYYY-MM-DD) on this resource. An example is available below:

```
{
    "resourceType": "Parameters",
    "parameter": [
        {
            "name": "operation",
            "part": [
                {
                    "name": "type",
                    "valueCode": "add"
                },
                {
                    "name": "path",
                    "valueString": "Task.executionPeriod"
                },
                {
                    "name": "name",
                    "valueString": "end"
                },
                {
                    "name": "value",
                    "valueDateTime": "{{EndExecutionDate}}"
                }
            ]
        }
    ]
}
```

Snippet 6: PATCH example on a BePerformerTask to finish an execution

8.4.2.2.5 Response

On successful update of the execution, the server sends back a HTTP 200. The response does not contain a body.

8.4.3 Interruption of an execution

A caregiver can interrupt his execution on a prescription.

8.4.3.1 Allowed roles

Caregiver

8.4.3.2 Endpoint

PATCH /Task/{executionID}

{executionID} being the ID of the BePerformerTask to be updated from a BeReferralServiceRequest.

8.4.3.3 Implemented rules

The interruption of an execution on an existing prescription is possible only if all following rules are respected:

- The caregiver is not suspended
- The BePerformerTask is in the ***status*** "in-progress".
- The information contained in the ***owner*** field matches with the information of the connected caregiver in the token.
- The business status of the prescription is "In Progress".
- No Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations are performed during the interruption of an execution, please refer to section 6.2.5.

8.4.3.4 Request

The operation is done through a PATCH on the BePerformerTask FHIR resource. It should only replace the status of this resource to the one “on-hold”. An example is available below:

```
{
    "resourceType": "Parameters",
    "parameter": [
        {
            "name": "operation",
            "part": [
                {
                    "name": "type",
                    "valueCode": "replace"
                },
                {
                    "name": "path",
                    "valueString": "Task.status"
                },
                {
                    "name": "value",
                    "valueString": "on-hold"
                }
            ]
        }
    ]
}
```

Snippet 7: PATCH example on a BePerformerTask to interrupt an execution

8.4.3.5 Response

On successful interruption of the execution, the server sends back a HTTP 200. This ID cannot be used anymore for further interactions with this resource.

8.4.4 Revoke an execution

A caregiver can revoke an execution on a prescription.

8.4.4.1 Allowed roles

Caregiver and Organization.

8.4.4.2 Endpoint

PATCH /Task/{executionID}

{executionID} being the ID of the BePerformerTask to be updated from a BeReferralServiceRequest.

8.4.4.3 Implemented rules

The revocation of an execution on an existing prescription is possible only if all following rules are respected:

- The caregiver is not suspended.
- The BePerformerTask is in the ***status*** “in-progress”.
- The information contained in the ***owner*** field must match with the information of the connected caregiver in the token.
- The business status of the prescription is “In Progress”.
- No Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations are performed during the revocation of an execution, please refer to section 6.2.5.

8.4.4.4 Request

The operation is done through a PATCH on the BePerformerTask resource. It should only replace the status of this resource to the one “revoked”. An example is available below:

```
{
    "resourceType": "Parameters",
    "parameter": [
        {
            "name": "operation",
            "part": [
                {
                    "name": "type",
                    "valueCode": "replace"
                },
                {
                    "name": "path",
                    "valueString": "Task.status"
                },
                {
                    "name": "value",
                    "valueString": "cancelled"
                }
            ]
        }
    ]
}
```

Snippet 8: PATCH example on a BePerformerTask to revoke an execution

8.4.4.5 Response

On successful revocation of the execution, the server sends back a HTTP 200. After this operation, the resource can be re-used by the caregiver.

8.5 Consultation of resources

8.5.1 Consultation of a BeReferralServiceRequest

This method allows the user to get information for one specific prescription by giving its reference ID.

It is also good to point out that after the creation of a prescription (BeReferralServiceRequest resource), in addition to the reference ID created for the prescription itself, a short code is also generated. The code is composed of 6 alphanumerical characters.

Therefore, the user can consult a BeReferralServiceRequest via the combination of the short code (=***identifier***) and patient SSIN (=***patient***).

If the user needs more resources around the BeReferralServiceRequest, like BePerformerTask, the recommended way is to use the $graph operation detailed in section 8.5.4. Multiple resources can be accessed with this $graph operation in one call to UHMEP API.

8.5.1.1 Allowed roles

Prescriber, Caregiver, Patient and Organization.

8.5.1.2 Endpoint

GET /ServiceRequest/{referenceId}

POST /ServiceRequest/_search (**for short code**)

8.5.1.3 Implemented rules

The consultation of a specific prescription is possible only if all the following points are respected:

- The referenceId given in the endpoint exists in UHMEP.
- When searching via Short Code + Patient SSIN, the system automatically replaces any letter 'O' or 'o' with the number '0' to prevent checksum failures.
- The prescription is in a status other than “Blacklisted”, or it is in “Blacklisted” status but has at least one execution associated with it.
- If the prescription status is "Cancelled" (revoked), it is **strictly restricted**. Caregivers will be blocked from accessing canceled prescriptions.
- (Prescriber) He is not suspended if he is not the prescriber of the prescription.
- (Caregiver) He is not suspended if he does not have any BePerformerTask linked to this prescription.
- (Patient) The prescription is for the patient requesting the prescription.
- Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations, please refer to section 6.2.5.

In case of a suspension:

- A prescriber will only be able to see a prescription created by him.
- A caregiver will only be able to see a prescription on which he is assigned or he works.

8.5.1.4 Request

In addition of the authorization and header described above, this request does not need a body.

8.5.1.5 Response

On successful request, the response will contain the requested prescription.

This prescription is represented in FHIR by a Belgian profile of the ServiceRequest resource.

8.5.2 Consultation of a list of BeReferralServiceRequests

When searching for data through UHMEP API, the user can use filters to narrow down the results to specific subsets of prescription and proposals that match their needs. The filters are applied to the search query in a logical AND manner, meaning that only the combinations of filters that match all the conditions specified will be returned in the search results.

Here are the key principles to keep in mind when using filters:

- **Filters are applied in a logical AND manner**: The search results are filtered by each individual filter, and only the records that match all the filters are included in the final set of results.
- **If no filters are specified, all search results will be returned**.
- **Filters can be combined in complex ways**: The user can combine multiple filters in a single query to create complex search criteria. For example, the user can filter by a specific patient and then further narrow down the results by a specific status.
- **If there is no prescription matching all filters used by the user, UHMEP will send back an empty list of prescriptions.**
- **If an unknown filter is provided, it is ignored**: UHMEP will use only filters that it understands.
- **If an unknown value is provided for a filter that has a FHIR defined list of defined values**: Since filters are used in a logical AND manner, it will return an empty list of results.

8.5.2.1 Allowed roles

Prescriber, Caregiver, Patient and Organization

8.5.2.2 Endpoint

POST /ServiceRequest /_search

8.5.2.3 Implemented rules

Each prescription or proposal that respect the requested search criteria will only be returned if it follows visibility rules of section 6.2 and 8.1.7.

8.5.2.4 Request

Here is a description of all filters available:

FHIR Filter	Value Required	Business Description
patient	Patient pseudonym in transit (domain UHMEP)	Filter by patient of the prescription
status	{status}	Filter by prescription status
statusReason	{statusReason}	Filter by additional info for prescription status
_has:Task:focus:status-_profile	{status}$Profile/be-referral-task	Filter by summary of executions status
requester	PractitionerRole/{Prescriber SSIN}-{Prescriber discipline}	Filter by prescriber of the prescription
_has:Task:focus:owner	PractitionerRole/{Caregiver SSIN}-{Caregiver discipline}	Filter by caregiver assigned to the prescription
_has:Task:focus:owner	Organization/{IdType}-{Id}-{OrgType}	Filter by organization assigned to the prescription
code	{code system}|{template code}	Filter on the template of the prescription
intent	{intent}	Filter between prescriptions and proposals
_count	Number	For pagination purpose, see description below
_offset	Number	For pagination purpose, see description below

Be aware that business statuses must be translated to FHIR statuses as referenced in section 6.3.2.1.

Those filters should be provided in the body of the POST request as *x-www-form-urlencoded* key-values.

Sorting

In UHMEP, ServiceRequests are sorted on their authoredOn fields when displaying a list. Upon searching for prescriptions/proposals using filters, the results are logically sorted from the newest on top, to the oldest at the bottom.

Pagination

While applying filters, the result may contain a large number of prescriptions. For performance reasons, UHMEP API uses a pagination.

_count

The _count filter is used to limit the number of search results returned by the API. It specifies the maximum number of results to return in a single page. For example, if the user sets _count=10, the API will return the first 10 matching records and omit the rest.

If this filter is not present, UHMEP will use the default and maximum allowed value that is currently set to 25. If the user requests more than this maximum, he will get the INVALID_COUNT_PARAMETER error.

_offset

The _offset filter is used to specify the starting point for the search results. It indicates the position of the first record to return in the result set. For example, if the user sets _offset=20, the API will start returning results from the 21st record onwards (i.e., skipping the first 20 records).

Together, _count and _offset can be used to implement pagination, allowing the user to navigate through large result sets in a controlled manner. For example:

- Set _count=10 and _offset=0 to retrieve the first 10 records.
- Set _count=10 and _offset=10 to retrieve the next 10 records, starting from the 11th record.
- And so on...

If this filter is not present, UHMEP will use the default value: 0

By using _count and _offset filters, the user can efficiently page through large result sets and retrieve only the data he needs.

Filter Access Control

FHIR Filter	Prescriber	Caregiver	Patient	Head Nurse	Admin
patient	Partially Mandatory	Partially Mandatory	Mandatory	Free	Free
status	Free	Free	Free	Free	Free
statusReason	Free	Free	Free	Free	Free
_has:Task:focus:status-_profile	Free	Free	Free	Free	Free
requester	Partially Mandatory + Partial Token Verification	Partially Mandatory + Partial Token Verification	Free	Free	Free
_has:Task:focus:owner	Partially Mandatory + Partial Token Verification	Partially Mandatory + Partial Token Verification	Free	Free	Free
_has:Task:focus:owner	Free	Free	Free	Mandatory	Mandatory
code	Free	Free	Free	Free	Free
intent	Free	Free	Free	Free	Free
_count	Free	Free	Free	Free	Free
_offset	Free	Free	Free	Free	Free

This table shows the specific rules depending on the role of the user accessing these search filters.

The following access control rules apply to each filter:

- **Free**: Users can choose to use or ignore this filter as needed.
- **Forbidden**: Users cannot use this filter. If they attempt to do so, they will receive an error message (SEARCH_CRITERIA_NOT_ALLOWED).
- **Mandatory**: Users must provide a value for this filter. If they do not, they will receive an error message (REQUIRED_SEARCH_CRITERIA_MISSING).
- **Partially Mandatory**: Users must use at least one of the *Partially Mandatory* filters. If they do not use any, they will receive an error message (REQUIRED_SEARCH_CRITERIA_MISSING).
- **Partially Mandatory + Partial Token Verification**: If the filter *patient* is **not** provided, then one of the Partial Token Verification should match with the token of the user (WRONG_REFERENCE_HEALTHCARE_PROFESSIONAL).
    - It implies that if the *patient* is provided, there should be **no** verification on the correspondence with the token content.

Example:

- If a Prescriber does a search without specifying a *patient,* he must:
    - Provide at least the *requester* or *_has:Task:focus:owner=PractitionerRole/....* filter. If not he will get (REQUIRED_SEARCH_CRITERIA_MISSING).
    - One of the *requesters* OR *_has:Task:focus:owner=PractitionerRole/....* filter should match his token. If not he will get (WRONG_REFERENCE_HEALTHCARE_PROFESSIONAL).

8.5.2.5 Response

On successful request, the response will contain a list of prescriptions corresponding to the filter(s) used.

The list is represented in FHIR by a Bundle resource.

Each prescription is represented in FHIR by a Belgian profile of the ServiceRequest resource.

8.5.3 Consultation of BeReferralTask and BePerformerTask

Alongside the direct consultation of FHIR resources using their specific IDs, it is also possible to do a search to receive Tasks linked on a BeReferralServiceRequest ID. It is possible to consult all BeReferralTask and BePerformerTask linked to a BeReferralServiceRequest but also do a filter to only get the BeReferralTask or only all BePerformerTask.

8.5.3.1 Consultation of all Tasks linked to a BeReferralServiceRequest

8.5.3.1.1 Allowed roles

Prescriber, Caregiver, Patient and Organization.

8.5.3.1.2 Endpoint

```
GET /Task?focus={BeReferralServiceRequestID}
```

8.5.3.1.3 Implemented rules

The consultation of a specific prescription is possible only if all rules described in section 8.5.1 are followed. Since Tasks linked to a BeReferralServiceRequest are part of the referral prescription, all same rules should also be applied.

8.5.3.1.4 Request

In addition of the authorization and header described above, this request does not need a body.

8.5.3.1.5 Response

A Bundle of type “searchset” will be returned on successful request. Entries will contain each BeReferralTask and BePerformerTask linked to the BeReferralServiceRequest

8.5.3.2 Consultation filter of Tasks linked to a BeReferralServiceRequest

8.5.3.2.1 Allowed roles

Prescriber, Caregiver, Patient and Organization.

8.5.3.2.2 Endpoint

```
GET /Task?focus={BeReferralServiceRequestID}&_profile={value}
```

Available "values" are described below:

value	Description
Profile/be-referral-task	Consult the BeReferralTask linked to the BeReferralServiceRequest
https://www.ehealth.fgov.be/standards/fhir/referral/StructureDefinition/be-referral-task	Consult the BeReferralTask linked to the BeReferralServiceRequest
Profile/be-performer-task	Consult all BePerformerTask linked to the BeReferralServiceRequest
https://www.ehealth.fgov.be/standards/fhir/referral/StructureDefinition/be-performer-task	Consult the BePerformerTask linked to the BeReferralServiceRequest

8.5.3.2.3 Implemented rules

The consultation of a specific prescription is possible only if all rules described in section 8.5.1 are followed. Since Tasks linked to a BeReferralServiceRequest are part of the referral prescriptions, all same rules should also be applied.

8.5.3.2.4 Request

In addition of the authorization and header described above, this request does not need a body.

8.5.3.2.5 Response

A Bundle of type "searchset" will be returned on successful request. Depending on the filter used, the Bundle will contain all BePerformerTask or the BeReferralTask linked to a BeReferralServiceRequest.

8.5.4 Consultation of a referral prescription (recommended)

All clinical information is stocked in the BeReferralServiceRequest, but other resources are often needed to have a global information. Every user that has the right to consult a prescription (see section 8.5) will also be able to ask for a Bundle that contains every resource that represents the global prescription. A visualization of resources that constitute a referral prescription is in Figure 4.

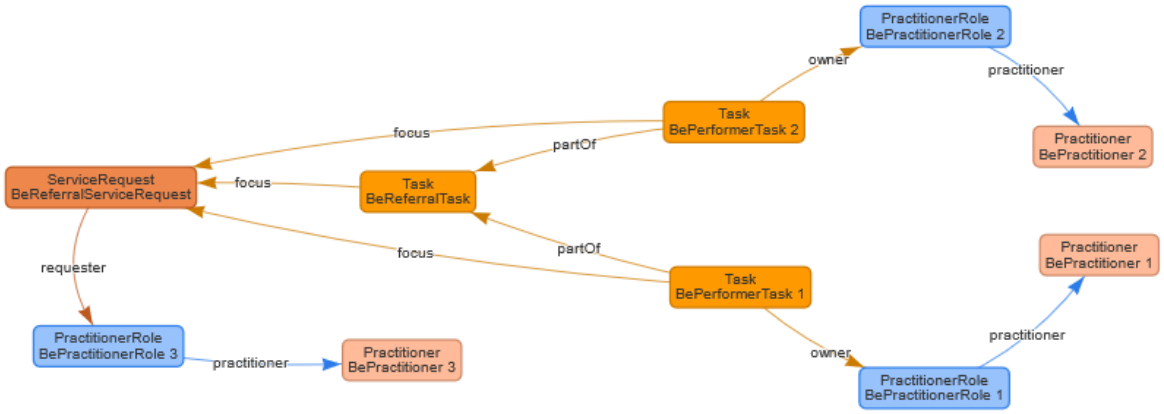


Figure 4: Example of FHIR resources that compose a referral prescription with two different executions on it

Users will have the possibility to get all these resources, with as many BePerformerTask/BePractitionerRole/BePractitioner as needed, in one call.

8.5.4.1.1 Allowed roles

Prescriber, Caregiver, Patient and Organization.

8.5.4.1.2 Endpoint

GET /ServiceRequest/{BeReferralServiceRequestID}/$graph?graph={GraphDefinitionURI}

GraphDefinitionURI available	Description
https://www.ehealth.fgov.be/standards/fhir/referral/GraphDefinition/full-referral-prescription	Entries will contain each BeReferralServiceRequest, BeReferralTask, BePerformerTask, BePractitionerRole and BePractitioner linked to the first BeReferralServiceRequest.

8.5.4.1.3 Implemented rules

The consultation of a specific prescription is possible only if all rules described in section 8.5.1 are followed. Since Tasks linked to a BeReferralServiceRequest are part of the referral prescription, all same rules should also be applied.

8.5.4.1.4 Request

In addition of the authorization and header described above, this request does not need a body.

8.5.4.1.5 Response

A Bundle of type “searchset” will be returned on successful request. Entries will contain each BeReferralServiceRequest, BeReferralTask, BePerformerTask, BePractitionerRole and BePractitioner linked to the first BeReferralServiceRequest.

For example, for the use case presented in Figure 4, the Bundle will contain 10 different entries.

8.6 Annex 81

A care proposal Annex 81 is a reverse process where a caregiver proposes a medication preparation to a prescriber. The prescriber then evaluates (approves or rejects) the proposal. If approved, UHMEP automatically generates a valid prescription from it.

8.6.1 Create an Annex 81 proposal

This method will allow the end user or an organization to create an Annex 81 proposal via UHMEP.

8.6.1.1 Allowed roles

Caregiver and Organization.

8.6.1.2 Endpoint

POST /ServiceRequest

8.6.1.3 Implemented rules

The following rules should be checked to create an Annex 81 proposal:

- The user connected is a caregiver
- The discipline of the caregiver has the right to “create a proposal” in the Access Matrix
- Use of the *intent* “proposal”

8.6.1.4 Request

The creation of the BeReferralServiceRequest resource must be done using the intent “proposal”. This is what makes it different from the creation of a prescription (which is a ServiceRequest with an intent “order”).

8.6.1.5 Response

On successful creation of the ServiceRequest resource, the server will respond with a HTTP 201. The ID of the proposal can be found in the location header. Additionally, the generated ID is now returned directly inside the JSON response payload.

8.6.2 Consult an Annex 81 proposal

This method allows the end user to consult an Annex 81 proposal by providing its RefID.

8.6.2.1 Allowed roles

Prescriber, Caregiver, Patient and Organization.

8.6.2.2 Endpoint

Same endpoint as for the consultation of a BeReferralServiceRequest (see 8.5.1)

8.6.2.3 Implemented rules

To consult an Annex 81 proposal, the following checks must be performed:

- The connected healthcare professional has the right to “consult a proposal”.
- The connected healthcare professional must not be suspended or he must be the creator or assigned on it.
- The connected patient must be the patient of the proposal that he wants to consult.
- Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations, please refer to section 6.2.5.

8.6.2.4 Request

The consultation of a proposal is possible by referencing its RefID.

8.6.2.5 Response

Upon a successful call, the server will respond with a 201 HTTP response and the user will be able to consult the BeReferralServiceRequest.

8.6.3 Assign an Annex 81 proposal

Allows to assign a specific prescriber to evaluate the proposal.

8.6.3.1 Allowed roles

Prescriber, Caregiver, Patient and Organization.

8.6.3.2 Endpoint

POST /Task

8.6.3.3 Implemented rules

- The target prescriber must have the "evaluate proposal" right.
- The user making the assignment must have the "assign proposal" right.
- The patient must be the subject of the Annex 81 proposal as referred in the “focus” field of the performerTask.
- Connected caregiver must not be suspended.
- The prescriber must be able to practice.
- The prescriber must not already be assigned.
- Proposal should be in the “ready” status.
- Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations, please refer to section 6.2.5.

8.6.3.4 Request

Just like for the assignation of a caregiver, the assignation is done through the creation of a BePerformerTask resource (see 8.3.1).

8.6.3.5 Response

On successful creation of the BePerformerTask resource, the server will respond with a HTTP 201. The ID of this newly created resource is accessible in the *Location* header. This ID can be used for further interactions with this resource.

8.6.4 Consult assignations on an Annex 81 proposal

This method allows a prescriber to consult the task created for the evaluation of an Annex 81 proposal.

8.6.4.1 Allowed roles

Prescriber, Caregiver, Patient and Organization.

8.6.4.2 Endpoint

GET /Task?focus={{proposalID}}

8.6.4.3 Implemented rules

Same as for Consultation of BeReferralTask and BePerformerTask (see 8.5.3).

8.6.4.4 Request

Same as for Consultation of BeReferralTask and BePerformerTask (see 8.5.3).

8.6.4.5 Response

Same as for Consultation of BeReferralTask and BePerformerTask (see 8.5.3).

8.6.5 Remove assignation on an Annex 81 proposal

Assignation on an Annex 81 proposal can be removed by the end users.

8.6.5.1 Allowed roles

Prescriber, Caregiver, Patient and Organization.

8.6.5.2 Endpoint

DELETE /Task/{executionID}

8.6.5.3 Implemented rules

- (Prescriber/Caregiver/Patient) The proposal referenced in the “focus” field is in the “ready” business status.
- (Prescriber/Caregiver/Patient) The execution of the prescriber (BePerformerTask) is in status “ready”.
- (Prescriber/Caregiver) The prescriber/caregiver is not suspended.
- (Patient) The patient is the same as the one in the proposal.
- (Prescriber) The connected prescriber can only remove his/her own bePerformerTask (owner).
- No Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations are performed during the removal of an assignation, please refer to section 6.2.5.

8.6.5.4 Request

This request does not need a body (see 8.3.2)

8.6.5.5 Response

On successful deletion of the resource, the server sends back a HTTP 204. After this operation, the resource is definitively deleted and cannot be used anymore.

8.6.6 Cancel an Annex 81 proposal

The end users are allowed to cancel an Annex 81 proposal when a medication preparation is not needed anymore.

8.6.6.1 Allowed roles

Caregiver, Patient and Organization.

8.6.6.2 Endpoint

PATCH /ServiceRequest/{{serviceRequestID}}

8.6.6.3 Implemented rules

- The proposal matches the following combination of prescription status and BeReferralTask status:
  - “active” – “ready”
- Connected caregiver is not suspended.
- Connected caregiver is the original creator of the proposal.
- Connected caregiver must have the right “cancel proposal”.
- Connected patient is the patient of the proposal.
- Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations, please refer to section 6.2.5.

8.6.6.4 Request

Same as for Cancellation of a prescription (see 8.2.2.4).

8.6.6.5 Response

Same as for Cancellation of a prescription (see 8.2.2.5).

8.6.7 Consult a list of proposals

As per the list of prescriptions, end users can retrieve a list of Annex 81 proposals.

That list only contains ServiceRequests with the intent “proposal”. The list is also sorted and a pagination is present (when applicable).

8.6.7.1 Allowed roles

Prescriber, Caregiver, Patient, Organization.

8.6.7.2 Endpoint

POST /ServiceRequest/_search

8.6.7.3 Implemented rules

Same as for the lists of prescriptions (BeReferralServiceRequest) (see 8.5.2). User should have the right to “consult_proposal”.

8.6.7.4 Request

Same as retreiving a list of BeReferralServiceRequests, with the exception that the body of the POST method may contain the intent “proposal” filter.

8.6.7.5 Response

Same as for 8.5.2.5: the response should contain a list of BeReferralServiceRequest with the intent “proposal”.

The same sorting logic applies for the proposal list consultation: the newest proposals are displayed on top of the page and the oldest at the bottom.

If applicable, a pagination is also put in place to avoid displaying too many results on the same page.

8.6.8 Evaluate (approve) an Annex 81 proposal

Allows an end user to approve the Annex 81 proposal, automatically converting it into an executable prescription.

8.6.8.1 Allowed roles

Prescriber.

8.6.8.2 Endpoint

POST /ServiceRequest/{{serviceRequestID}}/$approve

8.6.8.3 Implemented rules

- Annex 81 proposal in status “ready”.
- Connected prescriber has the “evaluate_proposal” right.
- Optional note provided in the operation
- Informed Consent, Therapeutic Relations and Therapeutic Exclusions validations, please refer to section 6.2.5.

Approving the proposal automatically:

1. Changes the proposal status to "completed".
2. Updates the evaluation BePerformerTask to "completed".
3. Automatically creates a valid Prescription and assigns the caregiver who authored the original proposal to it.

8.6.8.4 Request

The approval request can include an optional free-text note. **Because this is clinical free text, the note MUST be encrypted.**
The payload requires a specific FHIR Parameters body containing the pseudonymized key in the meta block and the encrypted JWE string.

For additional details, please refer to this page: https://build.fhir.org/ig/hl7-be/referral/branches/nursing-pilot/OperationDefinition-be-op-approve-annex81.html

8.6.8.5 Response

Upon successful approval, a 201 SUCCESS message should be sent. The ID created can then be used in the same way as the ID of a BeReferralServiceRequest resource with intent “order” (prescription).

8.6.9 Reject an Annex 81 proposal

Same as per the approval, end-users are also allowed to reject an Annex 81 proposal if it is deemed unnecessary/not required anymore.

8.6.9.1 Allowed roles

Prescriber.

8.6.9.2 Endpoint

POST /ServiceRequest/{{serviceRequestID}}/$reject

8.6.9.3 Implemented rules

The same set of rules apply as for the Annex 81 approval (see 8.6.8).

8.6.9.4 Request

Same as for the Annex 81 approval (see 8.6.8).

For additional details, please refer to this page: https://build.fhir.org/ig/hl7-be/referral/branches/nursing-pilot/OperationDefinition-be-op-reject-annex81.html

8.6.9.5 Response

SUCCESS 200 HTTP response expected upon successful Annex 81 proposal rejection.

8.7 Organization assignations

8.7.1 Assignation to an organization

Once a prescription has been created, the system allows users to assign a specific organization on it. Giving them an ease of access to this prescription and some access rights.

8.7.1.1 Allowed roles

Organization, Patient, Caregiver

8.7.1.2 Endpoint

This API request is the same as of an assignation of a caregiver. Please look on section 8.3.1.

8.7.1.3 Request

This operation is using a FHIR BeOrganizationTask resource, with the ***owner*** field as a reference to an organization, as described in section 6.3.4.

8.7.2 Remove assignation of an organization

This feature works as a DELETE on the BeOrganizationTask ID, it is the same feature as the “Remove a caregiver” in section 8.3.2.

8.7.3 Assignation of a caregiver working for an organization

A user being connected as working for an organization can assign a caregiver and specify that he/she is linked to the organization assignation.

8.7.3.1 Allowed roles

Organization

8.7.3.2 Endpoint

This API request is the same as of an assignation of a caregiver. Please look on section 8.3.1.

8.7.3.3 Request

The only field that changes from a simple assignation of a caregiver is the field ***focus*** that should be a reference to the existing BeOrganizationTask created in section 8.7.1. See Figure 2 for a visual representation of links between resources.

9 Error management

In the cycle of prescription, UHMEP API allows different type of requests and multiple different actors can interact with the API. If there is an error in the request, or if the action is forbidden by the system, UHMEP sends back an error. This section documents how errors will be represented by UHMEP. Alongside the correct HTTP code, UHMEP uses the FHIR resource OperationOutcome which is defined by FHIR to display the error.

An error can appear for different reasons. The errors returned by UHMEP are structured in a specific way to allow the end-user to quickly understand where the errors come from.

UHMEP uses issues with fields:

- ***severity***: defines if this is an "error" or a "warning". (https://hl7.org/fhir/r4b/valueset-issue-severity.html)
- ***code***: FHIR error type. UHMEP uses (business-rule, value, security, forbidden, conflict, processing). (https://hl7.org/fhir/r4b/valueset-issue-type.html)
- ***details.coding.system:*** The system where the error is defined. Errors thrown and defined by UHMEP are in the system "urn:uhmep:errors"
- ***details.coding.code***: The code that defines the error. In the document, it is referenced as the **UHMEP code**
- ***diagnostics***: Contains the description of the UHMEP code to help understanding. It should always be the same message for one UHMEP code. This message ends with an error id that can used for support purpose since this error is logged and more information can be retrieved by the UHMEP team through the error tracking system of UHMEP.

All error messages returned by UHMEP are listed in the Excel document "UHMEP_API_ErrorCodes.xlsx" alongside this cookbook.

10 Appendix A – Informed Consent, Therapeutic Relationships & Exclusions

10.1 Terminology

- **Required**: The specified condition must be met (e.g., an active therapeutic relationship must exist) for the action to proceed.
- **Forbidden**: The specified condition must not be met (e.g., the patient must not have an exclusion for the professional) for the action to proceed.
- **N/A**: This check is not performed for the given operation.

10.2 Example Walkthrough

Looking at the row for an "Assignation" where the connected user is not the original requester:

- UHMEP checks the Exclusion between the connected user and the patient.
- The **Forbidden** state means the action will only succeed if there is **NO** exclusion in place.
- If an exclusion **is** found, UHMEP will block the action. In this case, it will return an “EXCLUSION_EXISTS” error. See section 9 for error management.
- No checks are performed for Informed Consent or Therapeutic Relationship.

Summary Table

Operation	Connected User	Informed consent	Exclusion	Therapeutic relationship
Create a prescription	All users	No therapeutic relation, exclusion and informed consent checks are applicable for this action.		
Cancellation of a prescription/proposals	All users	No therapeutic relation, exclusion and informed consent checks are applicable for this action.		
Assignation of a healthcare professional/of an organization ***Additionally, the healthcare professional to be assigned should not be excluded by the patient to allow the assignation.***	• **Prescriptions**: Prescriber (also requester) • **Annex 81 proposals**: Nurse (also requester)	**N/A**	**N/A**	**N/A**
	• **Prescriptions:** Prescriber (not requester) • **Annex 81 proposals:** Nurse (not requester)	**N/A**	**Forbidden**	**N/A**
	• **Prescriptions**: Caregiver (assigned → _has:Task:focus:owner=himself) • **Annex 81 proposals**: Physician (assigned)	**N/A**	**Forbidden**	**N/A**
	• **Prescriptions:** Caregiver (not assigned) • **Annex 81 proposals:** Physician (not assigned)	**N/A**	**Forbidden**	**N/A**
	Patient	**N/A**	**N/A**	**N/A**
	Head Nurse	**N/A**	**Forbidden**	**N/A**
	Admin	**N/A**	**N/A**	**N/A**

Remove an assignation from a prescription/proposal (healthcare professional and organization)	All users	No therapeutic relation, exclusion and informed consent checks are applicable for this action.		
Start an execution	Caregiver (assigned → _has:Task:focus:owner= himself)	**N/A**	**N/A**	**N/A**
	Caregiver (not assigned)	**N/A**	**Forbidden**	**N/A**
Finish an execution	Caregiver connected is not yet assigned/has not yet started an execution (_has:Task:focus:owner != himself, only possible through a POST /Task with both start and end date of execution)	**N/A**	**Forbidden**	**N/A**
	Caregiver assigned	**N/A**	**N/A**	**N/A**
Cancellation of an execution	All users	No therapeutic relation, exclusion and informed consent checks are applicable for this action.		
Interruption of an execution	All users	No therapeutic relation, exclusion and informed consent checks are applicable for this action.		
Consultation of a list of prescriptions/proposal ***Rules are described in section 8.5.2 . No error will be returned, but each item (prescription/proposal) that does not follow those rules, will not be returned.***	• **Prescriptions**: Prescriber (also requester) • **Annex 81 proposals**: Nurse (also requester)	**N/A**	**N/A**	**N/A**
	• **Prescriptions:** Prescriber (not requester) • **Annex 81 proposals:** Nurse (not requester)	**Required**	**Forbidden**	**Required**
	• **Prescriptions**: Caregiver (assigned → _has:Task:focus:owner= himself)	**N/A**	**Forbidden**	**N/A**

	• **Annex 81 proposals**: Physician (assigned)			
	• **Prescriptions:** Caregiver (not assigned) • **Annex 81 proposals:** Physician (not assigned)	**Required**	**Forbidden**	**Required**
	Patient	**N/A**	**N/A**	**N/A**
	Head Nurse (organization is assigned → _has:Task:focus:owner= his organization)	**N/A**	**Forbidden**	**N/A**
	Head Nurse (organization not assigned)	**Required**	**Forbidden**	**Required**
	Admin (organization is assigned → _has:Task:focus:owner= his organization)	**N/A**	**N/A**	**N/A**
	Admin (organization not assigned)	**Required**	**N/A**	**N/A**
Consultation of a specific prescription ***This includes the consultation of Tasks, the Graph consultation and direct consultation of prescription/proposal. This is also valid for direct consultation via RefID and Short code + patient SSIN.***	• **Prescriptions**: Prescriber (also requester) • **Annex 81 proposals**: Nurse (also requester)	**N/A**	**N/A**	**N/A**
	• **Prescriptions:** Prescriber (not requester) • **Annex 81 proposals:** Nurse (not requester)	**N/A**	**Forbidden**	**N/A**
	• **Prescriptions**: Caregiver (assigned → _has:Task:focus:owner= himself) • **Annex 81 proposals**: Physician (assigned)	**N/A**	**Forbidden**	**N/A**
	• **Prescriptions:** Caregiver (not assigned)	**N/A**	**Forbidden**	**N/A**

	• **Annex 81 proposals:** Physician (not assigned)			
	Patient	**N/A**	**N/A**	**N/A**
	Head Nurse	**N/A**	**Forbidden**	**N/A**
	Admin	**N/A**	**N/A**	**N/A**
Create a proposal	All users	No therapeutic relation, exclusion and informed consent checks are applicable for this action.		
Approve/Reject an Annex81 proposal	Prescriber (assigned → _has:Task:focus:owner= himself)	**N/A**	**N/A**	**N/A**
	Prescriber (not assigned)	**N/A**	**Forbidden**	**N/A**